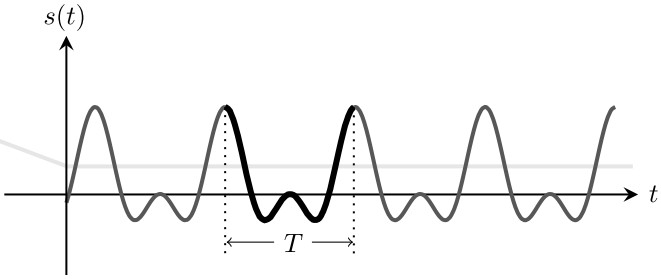
Filtrage

Signaux périodiques

Valeur moyenne

$$\langle s(t)\rangle = \frac{1}{T}\int_0^T s(t)\,dt$$



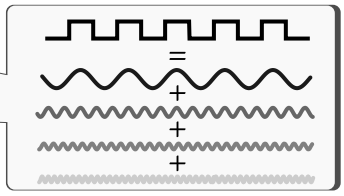
Valeur efficace

$$S = \sqrt{\frac{1}{T}\int_0^T s^2(t)\,dt}$$

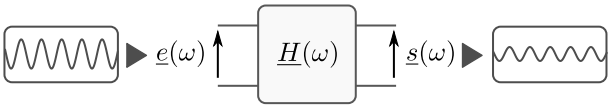
T	période
$f = \frac{1}{T}$	fréquence
$\omega = 2\pi f$	pulsation

Décomposition en série de Fourier

$$s(t) = \underbrace{c_0}_{\text{composante continue}} + \sum_{i=1}^{n} \underbrace{c_n \cos(n\omega t + \varphi_n)}_{\text{harmonique de rang } n}$$



Fonction de transfert



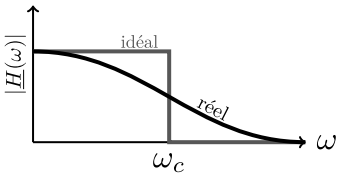
Fonction de transfert harmonique $\underline{H}(\omega) = \dfrac{\underline{s}(\omega)}{\underline{e}(\omega)}$

Gain $G = |\underline{H}(\omega)|$

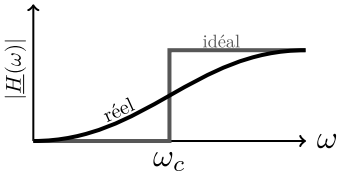
Déphasage $\varphi = \arg(\underline{H}(\omega))$

Gain en décibels $G_{dB} = 20\log(G)$

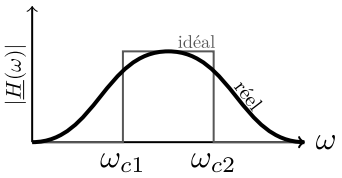
Types de filtres



passe-bas



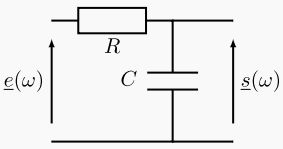
passe-haut



passe-bande

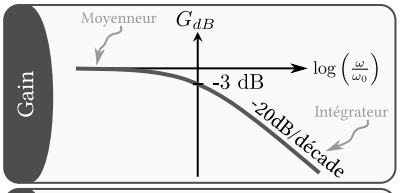
Filtre passe-bas

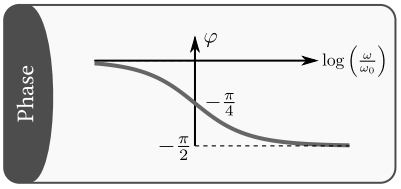
Filtre passe-bas d'ordre 1



Fonction de transfert

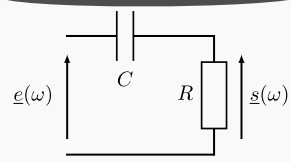
$$\underline{H}(\omega) = \frac{1}{1 + jRC\omega} = \frac{1}{1 + j\frac{\omega}{\omega_0}}.$$



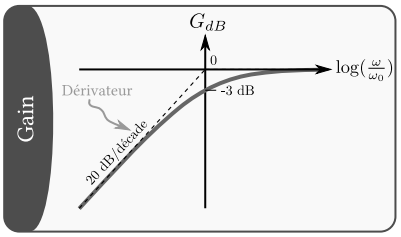
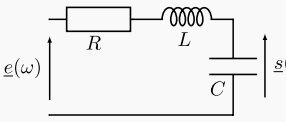
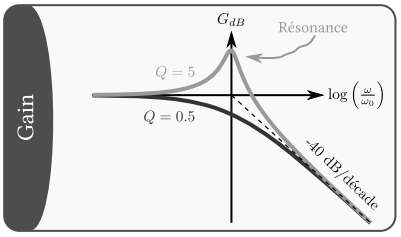


Autres filtres

Filtre pass-haut d'ordre 1



$$H(\omega) = \frac{1}{1 + (jRC\omega)^{-1}} = \frac{1}{1 + j\frac{\omega_0}{\omega}}$$



Filtre pass-base d'ordre 2